

Swapnadeep Sarkar

Bangalore, India
swapnamama90@gmail.com
9804176620

Summary

Results-oriented professional with a proven ability to manage and deliver comprehensive solutions across both business and technical domains. Skilled in analyzing and optimizing technical architectures, including AI-driven tools and integrations, to enhance performance and functionality. Adept at understanding customer journeys, driving innovation, and aligning technical solutions with business objectives to achieve impactful results.

Experience

Technical Consultant

Adobe

07/2022 - Present

- Built **MarTech and Growth Engineering** solutions across **AEP, AJO, Adobe Target, Analytics/CJA, Launch, and AEM EDS**, enabling personalization, experimentation, customer orchestration, and measurement.
- Explored **MCP integrations for AJO, AEP, CJA, and Target** with **ChatGPT, Claude, and Adobe Coworker**; Developed and evaluated internal MCP solutions for **email size optimization** and **Experience Decisioning**, achieving **35–40% reduction in email size** when applied to migration projects.
- Designed and shipped **Adobe Coworker skills for Experiment Accelerator**, including a **RAG-based Experiment Knowledge Base** with vector search and a conversational **Sample Size Calculator**, simplifying experiment discovery and planning for marketers.
- Developed **Python automation and REST API workflows** for **offer generation, migration, and lifecycle management**, reducing manual effort and accelerating bulk offer operations.
- Designed **AJO journeys and campaigns** using streaming/batch audiences, real-time events, APIs, conditional orchestration, and decisioning across **Email, Push, SMS, In-App, Web, and CBE**.
- Architected **AEP data solutions** covering XDM schemas, batch/streaming ingestion, identity stitching, profile unification, Data Distiller, audience segmentation, and downstream activation.
- Delivered **Adobe Target experimentation and personalization** using A/B testing, Experience Targeting, and Recommendations, contributing to **20–25% uplift in conversion/engagement metrics** across optimization use cases.
- Built a reusable **Adobe Target Recommendations component on AEM EDS** with configurable variations, enabling scalable deployment of recommendation experiences across web properties.
- Implemented **Offer/Experience Decisioning** with eligibility, ranking, and optimization strategies to deliver scalable, personalized experiences across customer journeys.
- Built frontend solutions using **JavaScript and React Native** (including AI-assisted development workflows), with reusable components and REST API integrations, improving application performance by **~20%** and engagement by up to **30%**.
- Integrated **Analytics/CJA and Launch** for web/mobile instrumentation, event tracking, campaign measurement, and experimentation analysis.
- Built validation and troubleshooting workflows across **APIs, events, profiles, journeys, and data pipelines**, improving production reliability and accelerating issue resolution.
- Collaborated with **engineers, architects, product managers, and marketing teams**, owning solutions from technical design through implementation, testing, deployment, and optimization.

UI/UX Design & Web Development Intern

IIFY Software

06/2020 - 09/2020

- Developed **responsive front-end** for two websites in the **Tourism and Events industries** using **HTML5, CSS3, and vanilla JavaScript** to create interactive components (e.g., dynamic sliders, form validations), resulting in improved

user engagement and streamlined navigation.

- **Designed and implemented wireframes and UI/UX mockups for a mobile application** tailored to military personnel facilities, ensuring intuitive navigation, adherence to accessibility standards (WCAG), and a consistent visual style across screens.
 - **Employed vanilla JavaScript to add dynamic functionality**, including interactive maps, modals, and real-time data displays. Optimized code for cross-browser compatibility, leveraged Chrome DevTools for debugging, and improved load times by minimizing DOM manipulations.
 - **Collaborated with designers and backend developers using Git/GitHub**, participated in code reviews to maintain code quality, and integrated RESTful API endpoints to deliver seamless data-driven features.
-

Skills

HTML, CSS, JavaScript, Bootstrap, JQuery, JSON, Python, SQL, Application Development, Adobe Target, Adobe Journey Optimizer, Adobe Experience Platform, Adobe Launch, Data Analysis and Machine Learning, GIT, JIRA, Devops, Problem-solving and Analytical Thinking, Software development, React Native, Google Firebase, Adobe Experience Manager (AEM), DOM Manipulation, QA & Debugging, Campaign Execution, Personalization, LLM Integration, Prompt Engineering, MCP (Model Context Protocol)

Education

M.Tech in Artificial Intelligence and Machine Learning

Birla Institute of Technology and Science • Pilani

03/2026

- Cumulative Grade Point Average (CGPA): 9.68
- Application development of different AI & ML modals.
- Developed an end-to-end AIML-driven customer intelligence framework integrating behavioral segmentation, segment-specific prediction models, LLM-guided use-case generation, and ROI simulation to operationalize unified customer data into actionable revenue strategies.

B.Tech in Electronics and Communication Engineering

Vellore Institute of Technology • Vellore

06/2022

- Cumulative Grade Point Average (CGPA): 8.6
 - Vice-President of IEEE - Circuits and Systems (CAS) at VIT, overseeing planning and execution of skill development workshops during GraVITas, managing technical workshops, handling event sponsorship, publicity efforts, and designing posters for flagship events
 - Published paper titled "Ultrasonic Sensor Based 3D Mapping and Localization" in IJCRT with an Impact Factor of 7.97
 - Contributed an article titled "ML-Based Finger-Vein Biometric Authentication and Hardware Implementation Strategies" in the book "Neuromorphic Computing Systems for Industry 4.0," Chapter 8 (pages 204-208)
-

Certificates

- Adobe Certified Expert - Adobe Journey Optimizer Developer from Adobe
 - Adobe Certified Master - Adobe Target Architect from Adobe
 - Adobe Target Optimization Specialist from Adobe
 - Python for Data Science and Machine Learning from Udemy
 - The Complete 2021 Web Development from Udemy
 - Software Engineering Virtual Experience of JPMorgan Chase
-

Awards

Consultant of the Quarter - Q3 - FY23 (Creative Catalyst), Award for "True essence" - Q4 - FY23

Social Links

Linkedin - www.linkedin.com/in/swapnadeep-sarkar; Github - www.github.com/Swapnadeep2K;
